

SOFTWARE ENGINEERING LAB 1

PROBLEM STATEMENT #32

Personal Subscription & Recurring Expense

Auditor

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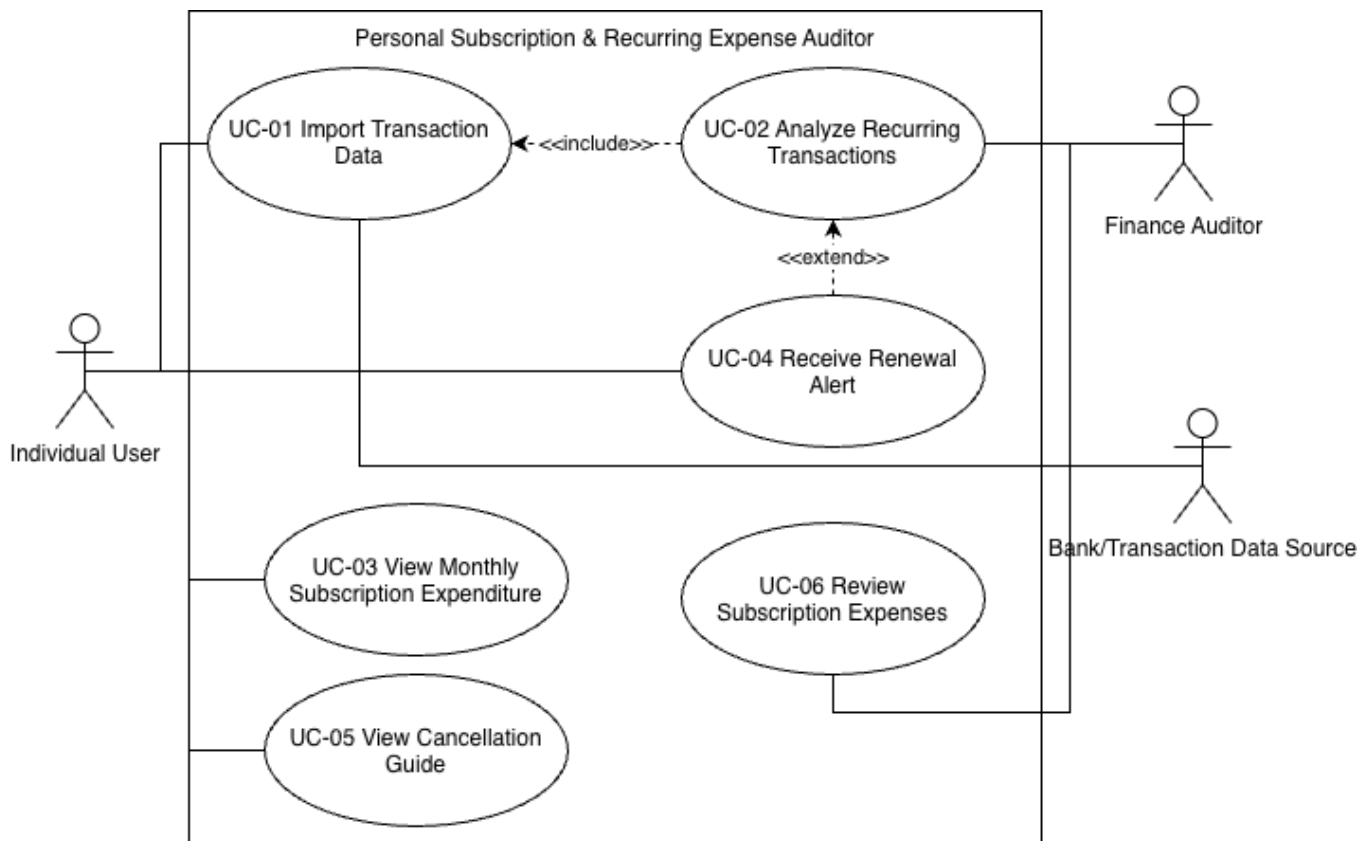
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REQUIREMENTS TABLE

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall analyze transaction logs to identify recurring billing patterns and project total monthly subscription expenditure.	High	Pass: Recurring service is identified and its renewal date is logged. Fail: A one-time payment is misclassified as a subscription.	Core functionality for identifying recurring expenses.
FR-002	Functional	The system shall alert the user before an identified subscription reaches its renewal date.	High	Pass: The user receives an alert before the recorded renewal date. Fail: No alert is generated for a subscription with a valid renewal date.	Helps users avoid unwanted or unexpected renewals.
FR-003	Functional	The system shall calculate and display the user's monthly recurring subscription expenditure.	High	Pass: The displayed monthly burn rate matches the sum of identified recurring subscription charges. Fail: The displayed total differs from the calculated amount.	Allows users to understand their recurring financial commitments.
FR-004	Functional	The system shall provide a cancellation guide for an identified subscription.	Medium	Pass: Selecting a subscription displays its applicable cancellation instructions. Fail: No cancellation guidance is available for a subscription that has a guide.	Helps users cancel unwanted subscriptions easily.
FR-005	Functional	The system shall allow the Finance Auditor to review identified recurring transactions and subscription expenditure.	Medium	Pass: The Finance Auditor can view identified recurring services and their expenditure. Fail: The auditor cannot access the required recurring-expense information.	Supports financial review and auditing of recurring expenses.

NFR-01	Nonfunctional	Financial transaction imports (CSV/OFX) must parse securely without storing raw banking credentials in cleartext.	High	Pass: CSV/OFX imports are processed without storing raw banking credentials in cleartext. Fail: Raw credentials are stored in cleartext.	Protects sensitive financial information.
NFR-02	Nonfunctional	The system shall generate subscription analysis results within 5 seconds for a transaction file containing up to 10,000 records.	High	Pass: Analysis completes within 5 seconds for a 10,000-record file. Fail: Analysis takes more than 5 seconds.	Ensures responsive analysis for realistic transaction volumes.

UML DIAGRAM



USE-CASE FLOW SPECIFICATION

Use Case ID: UC-02

Use Case Name: Analyze Recurring Transactions

Primary Actor: Individual User

Supporting Actor: Bank/Transaction Data Source

Preconditions

1. The user has access to the application.
2. Transaction data has been imported into the system.
3. The imported transaction data is available for analysis.

Postconditions

1. Recurring transactions are identified.
2. Subscription renewal dates are recorded where available.
3. The estimated monthly subscription expenditure is calculated and displayed.

Main Success Scenario

1. The Individual User imports transaction data into the system.
2. The system validates the imported transaction data.
3. The system analyzes the transaction records for recurring billing patterns.
4. The system identifies transactions that represent recurring subscriptions.
5. The system records the identified subscription details and renewal dates.
6. The system calculates the total monthly subscription expenditure.
7. The system displays the identified subscriptions and monthly expenditure to the user.
8. The use case ends successfully.

Alternate Flow: A1. Invalid or Unreadable Transaction Data

1. At Step 2, the system detects that the imported transaction data is invalid or cannot be parsed.
2. The system displays an error message to the user.
3. The user is prompted to provide a valid transaction file.
4. The use case resumes from Step 1 of the Main Success Scenario.